KARTHICK A

Embedded Product Engineer

Automotive ECU Electronics · Custom PCB Design · Embedded C/C++ · Industrial IoT

- **2+ Years** | Embedded Product Development
  Concept → design → field validation
- **Dual-Fuel (LPG/CNG) ECU**
  Engine sensing + injector drive — validated on a running engine
- **End-to-End Hardware Ownership**
  Schematic → PCB layout → bring-up → validation
- **Commercial Deployments**
  Delphi TVS | Indo Tech Transformers

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English (Professional) · Tamil (Native)

SKILLS

Engine & ECU Electronics

- **Dual-fuel ECU (LPG/CNG)** · injector drivers · fuel control logic
- **Engine sensing: MAP, EGT, RPM** · sensor calibration · CAN interfacing

Firmware

- **Embedded C / C++** · interrupt-driven · state machines · drivers: GPIO, ADC, PWM, timers
- **PlatformIO** · STM32CubeIDE · Arduino IDE · Git
- Learning: FreeRTOS · STM32 bare-metal · Linux (Debian)

Microcontrollers

- **STM32 F1/F4** · ESP32 / S3 · ESP8266 · Arduino Mega/UNO
- **ATtiny 3216 / 2313** — bare-IC design, UPDI

Hardware & PCB

- **Schematic capture** · PCB layout · BOM · bring-up · validation
- **MOSFET driver circuits** · linear supplies · buck converters
- **Flyback / inductive-load** · reverse-polarity protection
- **Altium Designer** · KiCad · EasyEDA · Proteus

Protocols & Data

- **CAN** · RS485 · UART · SPI · I2C · LoRa · HTTP/JSON · SD/SPIFFS logging

Test & Debug

- **Oscilloscope** · multimeter · bench power supply
- **ST-Link SWD** · UPDI programming · USB-UART
- **Custom PC-side UART capture tool** · board-level fault analysis

EDUCATION

M.E. Embedded System Technologies
Sri Sai Ram Engg. College — CGPA 7.82

B.E. Electrical & Electronics Engineering
Sri Sai Ram Engg. College — 75%, First Class

AWARDS & CERTIFICATIONS

Youth Co:Lab (UNDP) — Top-20 Finalist
SIH 2025 Mentor · DPIIT Startup · IVP Grant
Certs: IIT Bombay · IIRS/ISRO (5) · IEEE (3) · NPTEL (2)

WORK EXPERIENCE

Embedded Product Engineer

Spark Invotech Pvt Ltd, Chennai

2024 – Present | 2+ years | DPIIT-recognized embedded systems company I co-founded

- Architected and delivered commercial embedded products end-to-end — requirements, hardware architecture, schematic, PCB layout, firmware, bring-up, validation and customer deployment.
- Led dual-fuel ECU and injector-driver development for IC engines; verified V1 on a running marine engine under operating load.
- Commissioned industrial monitoring electronics for **Delphi TVS** (diesel fuel-injection) and **Indo Tech Transformers Ltd** (power equipment) — on-site validation and technical documentation.

R&D Engineer Intern

AbhiVijay Technologies, Chennai

Jan 2024 – Apr 2024

- Developed ECU and OCU hardware — R&D, hardware design, testing and system-integration validation for automotive electronic control units.

KEY PROJECTS

Dual-Fuel Engine Control Unit (LPG/CNG)

Indigenous low-cost ECU — Spark Invotech

- Engineered an indigenous low-cost ECU architecture converting IC engines to dual-fuel LPG/CNG, including a marine engine retrofit.
- Integrated real-time engine sensing (MAP, EGT, engine speed) via precision ADC with in-house calibration across firmware versions; implemented fuel and injector control logic.

Tech: STM32, ESP32, Arduino Mega, ADS1115, MAP/EGT/RPM sensors, CAN module, Embedded C, STM32CubeIDE

Acoustic Vehicle Alerting System (AVAS) for EVs

Electric vehicle pedestrian-safety prototype

- Prototyped a speed-dependent acoustic warning system on ESP32-S3 — tone and volume rise with vehicle speed, extending engine-sensing work into EV safety hardware.

Tech: ESP32-S3, PWM audio generation, analog speed input, Embedded C/C++, PlatformIO

Injector Multiplier

ECU channel-expansion module

- Designed a power-electronics module multiplying 4 ECU injector channels to drive up to **16 injectors**, letting large multi-cylinder engines run on a low-channel, low-cost ECU.

Tech: 12 V / 8 A peak-per-channel driver stages, MOSFET switching, flyback and reverse-polarity protection, custom PCB

Oil Control Unit (OCU)

Electronic oil pump controller

- Designed, manufactured and verified two hardware versions on custom PCBs for regulated oil-pump delivery — ESP8266, PWM output, MOSFET/TIP122 power stage with protection.

Transformer Condition Monitoring System

Indo Tech Transformers Ltd — power equipment manufacturer · POC completed

- Developed an IoT logger acquiring pressure, voltage, current, temperature and running hours with RTC timestamping, local storage and cloud sync.

Tech: ESP, PT100, ZMPT101B, SCT013, RTC, SD logging, HTTP/JSON cloud APIs, P10 LED display

Cold Chamber Monitoring System

Delphi TVS — diesel fuel-injection systems manufacturer · delivered

- Architected a 3-node distributed data logger for temperature profiling (**+50 °C to −20 °C**) with a command/ACK protocol, 3-retry logic and session persistence across power cycles; validated on-site before handover.

Tech: ESP32, RS485 half-duplex (DE/RE), LoRa E32 (~200 m as deployed), SPIFFS logging, Python PC interface